HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Implemented and scaled **Python**-based training pipelines for large transformer and LLM models, covering data ingestion, preprocessing, and distributed training phases.
- Built and optimized **inference services** for low-latency, high-reliability customer experiences, incorporating autoscaling, routing, and fallback mechanisms.
- Worked on **GPU-level performance tuning**, improving kernel efficiency and identifying bottlenecks across the training and inference stack to enhance **CUDA** and **Triton** utilization.
- Collaborated closely with ML scientists to implement and productionize **cutting-edge training and inference methods**, ensuring seamless integration into the existing AI platform.
- Played an active role in mentoring junior engineers on best practices for **AI infrastructure development** and operational excellence.
- Drove improvements in technical standards and operational excellence across Fin's AI platform, focusing on **reliability** and **scalability** for large-scale model serving.
- Containerized AI model serving components using **Docker** and deployed them to a **Kubernetes** cluster on **AWS**, managing configurations and resource allocation.

SKILLS

AI/ML Infrastructure – Model Training, Model Inference, LLMs, Transformers, GPU Performance, CUDA, Triton, MLOps, AI Agent

Programming Languages – Python, Java, SQL, R

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), Kubernetes, Docker

Data Processing & ETL – Pandas, NumPy, Data Preprocessing, Data Ingestion

Software Development – REST APIs, Spring Boot, Microservices, CI/CD, Jenkins, Git, Code Reviews

Observability & Monitoring – Splunk, Dynatrace, CloudWatch

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

Distributed LLM Training Pipeline

- Engineered a **Python**-based distributed training pipeline for large language models on **AWS** using libraries like **PyTorch** and **Horovod**.
- Implemented data preprocessing steps for a multi-terabyte dataset, transforming raw text into tokenized sequences suitable for model consumption, processing approximately **5,000 records per second**.
- Integrated monitoring and logging using **CloudWatch** to track training progress, resource utilization, and identify potential bottlenecks in the distributed training environment.

Low-Latency Model Inference Service

- Developed a high-throughput, **low-latency inference service** using **FastAPI** and **Docker** to serve predictions from trained AI models.
- Optimized model loading and inference execution paths to achieve a p95 latency of **180ms** for customer-facing requests.
- Deployed the service to a **Kubernetes** cluster, configuring autoscaling policies to handle fluctuating traffic patterns and ensure service availability.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)